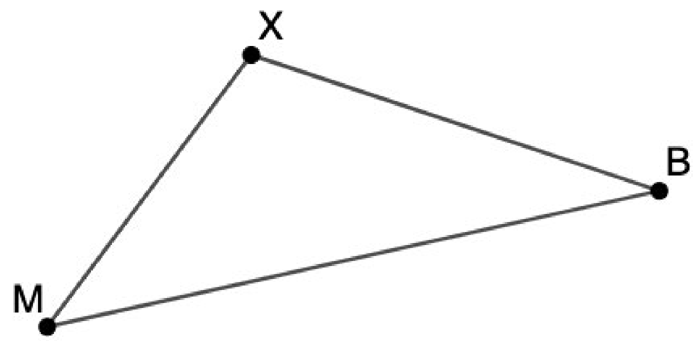
Geometry
Unit 11
Lesson 5 Practice

Name Answer Key______________
Date__________________
Period__________

ΔMXB is shown where $m\angle M = (5x)°$, $m\angle X = (15x - 10)°$, and $m\angle B = (5x - 10)°$.



1. Find the value of x.

   $5x + 15 - 10 + 5x - 10 = 180$
   $25x - 20 = 180$
   $x = 8$

2. What is the $m\angle M$?

   $m\angle M = 5(8) = 40°$

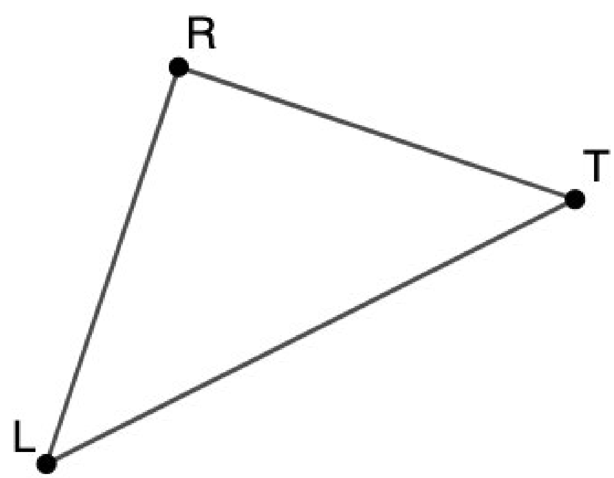
3. What is the $m\angle X$?

   $m\angle X = 15(8) - 10 = 110°$

4. What is the $m\angle B$?

   $m\angle B = 5(8) - 10 = 30°$

ΔLRT is shown, where $m\angle L = m\angle T = (2x + 6)°$ and $m\angle R = (4x + 12)°$. Figure is not drawn to scale.



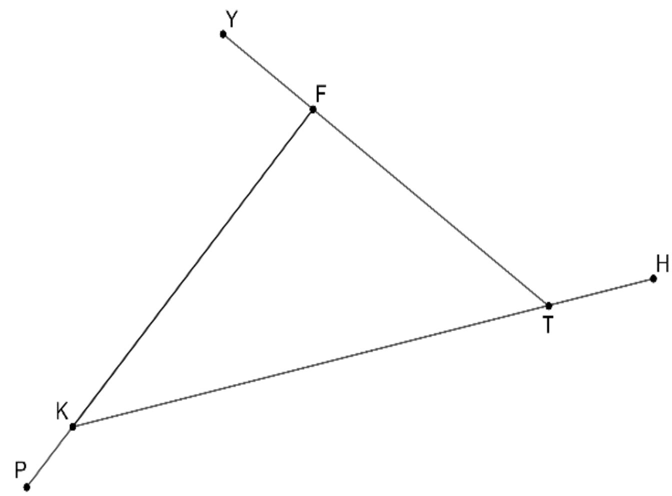
5. Find the value of x.

   $2(2x + 6) + 4x + 12 = 180$
   $8x + 24 = 180$
   $x = 19.5$

6. What is the $m\angle R$?

   $m\angle R = 4(19.5) + 12 = 90°$

ΔFKT is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles with $m\angle FTH = (12x + 25)°$, $m\angle YFK = 95°$ and $m\angle FKT = 24°$.



7. Find the value of x.

$12x + 25 + 95 + (180 - 24) = 360$
$12x + 276 = 360$
$12x = 84$
$x = 7$

8. What is $m\angle KTF$?

$m\angle KTF = 180 - (12 \cdot 7 + 25) = 180 - 109 = 71°$

9. What is $m\angle PKT$?

$m\angle PKT = 180 - 24 = 156°$

10. What is $m\angle KFT$?

$m\angle KFT = 85°$